

1. COMPUTING WITH REAL NUMBERS

Floating-point system

Normalized binary FPN: $\text{fl}(x) = \pm(1.d_1d_2\dots d_p)_2\,2^e$, $d_i \in \{0,1\}$, $e_{\min} \leq e \leq e_{\max}$. IEEE double: sign 1 bit, p=52 fraction bits, 11 exponent bits; normal $e \in [-1022, 1023]$. $\epsilon = 2^{-52} \approx 2.22 \times 10^{-16}$ (gap from 1 to next FPN); $e_i = 2^m e$ ($2^m \leq i < 2^{m+1}$), $x_{\max} = (2 - \epsilon)2^{1023}$, $x_{\min}^{\text{normal}} = 2^{-1022}$. Below: subnormal, then underflow to 0; overflow $\rightarrow \pm\infty$; undefined $\rightarrow \text{NaN}$. ***FPN arithmetic is not associative; avoid exact equality tests.*** **Errors, accuracy, stability** For approximation $\tilde{x}$ to x : $e_a = |\tilde{x} - x|$, $e_r = \frac{|\tilde{x} - x|}{|x|}$ (***$x \neq 0$***).

Significant decimal figures: largest p satisfying $e_r < \frac{1}{2}10^{-p}$. Storage rounding error $\leq \epsilon|x|$. **Catastrophic cancellation:** subtracting computed nonzero $\tilde{x}, \tilde{y}$ when $x \approx y$ can greatly amplify relative error. Quadratic $ax^2 + bx + c = 0$: $q = -\frac{1}{2}(b + \text{sign}(b)\sqrt{b^2 - 4ac})$, $x_1 = \frac{q}{a}$, $x_2 = \frac{c}{q}$. ***Require $a \neq 0$, real-root form $b^2 - 4ac \geq 0$, $q \neq 0$ for c/q .***

Cost and storage

1 bit $\in \{0,1\}$; 1 byte = 8 bits; KiB = 2^{10} bytes, MiB = 2^{20} , GiB = 2^{30} . Signed b-bit integer range $[-2^{b-1}, 2^{b-1} - 1]$. 1 flop = one FPN +, −, ×, /. Speed = (cycles/s)/(cores)(flops/cycle); time = flops/speed. memory($n \times n$) = $8n^2$ bytes, flops(AB) = $n^2(2n - 1) \approx 2n^3$.

2. NORMS, EQUALITY, SENSITIVITY

Vector norms and error

For $x \in \mathbb{R}^n$, $p \geq 1$: $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}$; $\|x\|_1 = \sum_i |x_i|$, $\|x\|_2 = \sqrt{x^T x}$, $\|x\|_\infty = \max_i |x_i|$. Norm axioms: positivity/definiteness, triangle inequality, absolute homogeneity. $x = y \iff \|x - y\| = 0$, $\|x - y\| \leq \tau(e, n, \|x\|, \|y\|)$. $e_a = \|\tilde{x} - x\|$, $e_r = \frac{\|\tilde{x} - x\|}{\|x\|}$ (***$x \neq 0$***); $\frac{\|\tilde{x} - x\|_\infty}{\|x\|_\infty} < \frac{1}{2}10^{-k}$, $k = \lfloor -\log_{10}(2e_r, \infty) \rfloor$.

Matrix norms

Subordinate norm: $\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}$; $\|Ax\|_p \leq \|A\|_p \|x\|_p$, $\|AB\|_p \leq \|A\|_p \|B\|_p$. $\|A\|_1 = \max_j \sum_i a_{ij}$, $\|A\|_\infty = \max_i \sum_j a_{ij}$, $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sigma_1$; Frobenius $\|A\|_F = \sqrt{\sum_{ij} |a_{ij}|^2}$ (***not generally $\|A\|_2$***).

Conditioning of $Ax = b$

For nonsingular A: $\kappa(A) = \|A\| \|A^{-1}\| \geq 1$; rcond = $1/\kappa$. $\kappa_2(A) = \frac{\max_i |x_i|}{\min_i |x_i|}$ ($A = A^T$), $\kappa_2(A) = \frac{\sigma_1}{\sigma_n}$ ($\text{rank}(A) = n$). $(A + \delta A)(x + \delta x) = b + \delta b$, $\|(A + \delta A)^{-1}\| \leq \alpha \|A^{-1}\|$; $A + \delta A$ nonsingular, $\alpha > 1$. $\rho_A = \frac{\|\delta A\|}{\|A\|}$, $\rho_b = \frac{\|\delta b\|}{\|b\|}$, $\frac{\|\delta x\|}{\|x\|} \leq \alpha \kappa(A)(\rho_A + \rho_b)$. Large $\kappa \Rightarrow$ sensitive/ill-conditioned; $\kappa \geq 1/\epsilon$ can destroy machine-precision information.

3. MATRIX STRUCTURE & FACTORISATIONS

Special matrices

$C_{ij} = E[(X_i - \mu_i)(X_j - \mu_j)]$, $C_{ii} = \sigma_i^2$; C symmetric PSD. $K_{ij} = \frac{C_{ij}}{\sigma_i \sigma_j}$, $K = \Sigma^{-1} C \Sigma^{-1}$, $K_{ii} = 1$, $|K_{ij}| \leq 1$ (***$\sigma_i, \sigma_j > 0$***). Upper/lower triangular: $a_{ij} = 0$ for $i > j$ / $i < j$. $\det A = \prod_i a_{ii}$, A nonsingular $\iff a_{ii} \neq 0 \, \forall i$. Unit triangular: $a_{ii} = 1$. Triangular solve $\approx n^2$ flops. $a_{ij} = 0$ if $i - j > m_\ell$ or $j - i > m_u$; total bandwidth $m_\ell + m_u + 1$; $\text{nnz} \leq (1 + m_\ell + m_u)n - m_\ell(m_\ell + 1)/2 - m_u(m_u + 1)/2$. Tridiagonal: $m_\ell = m_u = 1$. Toeplitz: $a_{ij} = \alpha_{j-i}$. Symmetric $A^T = A$; skew $A^T = -A$. $A = \frac{A + A^T}{2} + \frac{A - A^T}{2}$. Definiteness

by $x^T A x$: PD > 0 , PSD ≥ 0 , ND < 0 , NSD ≤ 0 ; for real symmetric A, signs of eigenvalues give type. Strict row diagonal dominance: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$; symmetric + strict dominance $\Rightarrow$ PD. $Q^T Q = I$, $Q^{-1} = Q^T$, $\kappa_2(Q) = 1$, $\|Qx\|_2 = \|x\|_2$. Permutation matrices are orthogonal.

LU and Cholesky

LU: $A = LU$ with L unit lower, U upper; when it exists for nonsingular A it is unique. In general partial pivoting gives $PA = LU$; $|\ell_{ij}| \leq 1$. Factor cost $\approx 2n^3/3$; two triangular solves $\approx 2n^2$. ***Require nonsingular A for a unique solve; do not form A^{-1}*** . Cholesky: $A = R^T R$, R upper. ***Require real symmetric positive-definite A***; factor cost $n^3/3$, solves $2n^2$; no row pivoting.

QR

For $A \in \mathbb{R}^{m \times n}$, $m \geq n$, $\text{rank}(A) = n$: $A = Q \begin{bmatrix} R \\ 0 \end{bmatrix} = YR$, $Q = [Y \, Z]$, $Q^T Q = I$; R nonsingular upper. Cost $\approx 2mn^2 - (2/3)n^3$. If rank deficient: $AP = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$, column pivoting, $|r_{11}| \geq \dots \geq |r_{nn}|$; small diagonal entries indicate numerical rank deficiency.

SVD

$A \in \mathbb{R}^{m \times n}$, $m \geq n$: $A = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^T$; U, V orthogonal, $\Sigma = \text{diag}(\sigma_1 \geq \dots \geq \sigma_n \geq 0)$. $\text{rank}(A) = \text{number of positive } \sigma_i$; $A = \sum \sigma_j u_j v_j^T$, $A v_i = \sigma_i u_i$, $u_i^T A = \sigma_i v_i^T$. Full rank: $\kappa_2(A) = \sigma_1/\sigma_n$. Numerical rank tolerance $\approx \max(m, n)\epsilon \sigma_1$.

4. PARAMETRIC DATA FITTING

Residual model

Data (t_i, y_i) , $i = 1, \dots, m$; parameters $x \in \mathbb{R}^n$, ***$m \geq n$*** . $r_i(x) = \phi(x; t_i) - y_i$, $x^* = \arg \min_x \|r(x)\|_p$. Interpolation $\Leftrightarrow r = 0$ (***appropriate only for highly accurate data***). Linear-in-parameters: $\phi(x; t) = \sum_j x_j \phi_j(t)$, $A_{ij} = \phi_j(t_i)$, $r = Ax - y$.

Linear least squares (p=2)

Normal equations: $A^T A x^* = A^T y$; residual orthogonal to $\text{col}(A)$. $A^T A$ PSD, and PD iff A has full column rank. ***Avoid forming $A^T A$ when conditioning matters:*** $\kappa_2(A^T A) = \kappa_2(A)^2$. Pivoted QR $AP = Q[R; 0]$, $Q = [Y \, Z]$: $R(P^T x^*) = Y^T y$. SVD $A = U[\Sigma; 0]V^T$, $U = [Y \, Z]$: $x^* = V\Sigma^{-1}Y^T y$, $x_{\min}^* = V\Sigma^+ Y^T y$. Use $\Sigma^+ = \text{diag}(1/\sigma_1, \dots, 1/\sigma_r, 0, \dots)$ for $\text{rank } r < n$.

Choice of norm / transformation

p=2: suited to independent $N(0, \sigma^2)$ errors with common variance. p=1: robust to outliers, $\min \sum v_i$: $-v_i \leq r_i \leq v_i$, $v_i \geq 0$. p= ∞ : $\min v$: $-v \leq r_i \leq v$, $v \geq 0$. Linear r gives linear programs. Log-linearisation: products/exponentials/powers may be transformed. ***Log transform requires every logged quantity positive and any shifted base $t - \beta_j > 0$*** .

5. CUBIC SPLINES & SPARSE MATRICES

Polynomials and interpolation

$p(t) = \sum_{k=0}^n a_k t^k = a_0 + \ell_1 a_1 + \ell_2 a_2 + \dots + \ell_n a_n$ (Horner form uses n multiplies + n adds. Weierstrass: for continuous f on closed $[a, b]$, $\forall \epsilon > 0$ polynomial p with $|f - p| < \epsilon$ on $[a, b]$. Polynomial interpolation at distinct $t_0, \dots, t_n$: Vandermonde $Ax = y$, $A_{ij} = t_i^j$. ***High-degree equally spaced interpolation can be ill-conditioned and oscillatory; use low-degree pieces.***

Cubic spline

Knots $a = t_0 < \dots < t_n = b$. $s(t) = s_i(t)$ on $[t_{i-1}, t_i]$, degree(s_i) ≤ 3 , and $s \in C^2[a, b]$. Interpolation $s_i(t) = y_i$; at interior knots $s_i(t) = s_{i+1}(t_i)$, $s_i'(t) = s_{i+1}'(t_i)$, $s_i''(t) = s_{i+1}''(t_i)$. Two endpoint conditions required: natural: $s''(a) = s''(b) = 0$; clamped: $s'(a), s'(b)$; not-a-knot: $s_1'''(t_1) = s_2'''(t_1)$, $s_{n-1}'''(t_{n-1}) = s_n'''(t_{n-1})$. Uniform $h = t_i - t_{i-1}$, $d_i = (y_i - y_{i-1})/h$, $M_i = s''(t_i)$. For $t \in [t_{i-1}, t_i]$: $s_i(t) = \frac{M_{i-1}(t_i - t)^3}{6h} + \frac{M_i(t - t_{i-1})^3}{6h} + \left(\frac{y_{i-1}}{h} - \frac{M_{i-1}h}{6}\right)(t_i - t) + \left(\frac{y_i}{h} - \frac{M_ih}{6}\right)(t - t_{i-1})$. Interior system: $\frac{h}{6}M_{i-1} + \frac{2h}{3}M_i + \frac{h}{6}M_{i+1} = d_{i+1} - d_i$, $i = 1, \dots, n - 1$, plus endpoint equations. Natural: $M_0 = M_n = 0$; coefficient matrix symmetric tridiagonal diagonally dominant PD; Thomas solve $O(n)$.

5. CUBIC SPLINES & SPARSE MATRICES

(CONT.)

Sparse structure

Sparsity = #zeros/(mn); density = nnz/(mn). Diagonal density $\approx 1/n$; tridiagonal $\approx 3/n$. Hidden sparse product: $(I + BB^T)x = x + B(B^T x)$, $B \in \mathbb{R}^{n \times r}$, $r \ll n$. Store/operate on nonzeros. Fill-in = new nonzeros during factorisation; PAQ = LU can reduce fill-in. For banded $A = LU$ without disruptive pivoting, L/U retain lower/upper bandwidth; SPD $A = R^T R$ gives R same upper bandwidth.

1-D boundary-value finite difference

$-u'' = f$ on $(0, 1)$, $u(0) = \alpha$, $u(1) = \beta$, $x_j = j\Delta x$, $\Delta x = \frac{1}{n+1}$. $-\frac{U_{j+1} - 2U_j + U_{j-1}}{(\Delta x)^2} = f(x_j)$, $j = 1, \dots, n$; $U_0 = \alpha$, $U_{n+1} = \beta$; error $O((\Delta x)^2)$ under sufficient smoothness. Matrix tridiagonal Toeplitz SPD with diagonal 2 and off-diagonal -1 ; RHS endpoints add α, β .

6. FOURIER ANALYSIS & FFT

Periodic and trigonometric form

$g(t + T) = g(t)$, $f = \frac{1}{T}$, $\cos(\omega t)$: $T = \frac{2\pi}{\omega}$. Harmonic k: frequency k/T , angular frequency $2\pi k/T$. $g_n(t) = a_0 + \sum_{k=1}^n \left[a_k \cos\left(\frac{2\pi k t}{T}\right) + b_k \sin\left(\frac{2\pi k t}{T}\right) \right]$. Over any full period (shown on $[-T/2, T/2]$): $a_0 = \frac{1}{T} \int_{-T/2}^{T/2} g(t) dt$; $a_k = \frac{2}{T} \int_{-T/2}^{T/2} g(t) \cos\left(\frac{2\pi k t}{T}\right) dt$; $b_k = \frac{2}{T} \int_{-T/2}^{T/2} g(t) \sin\left(\frac{2\pi k t}{T}\right) dt$. Orthogonality: equal nonzero sine/sine or cosine/cosine integral = $T/2$; unequal or sine/cosine = 0.

If periodic g is piecewise differentiable: Fourier series converges to g at continuity points and to $[g(t+) + g(t-)]/2$ at jumps. Gibbs oscillations persist near jumps.

Complex Fourier series

$g(t) = \sum_{k \in \mathbb{Z}} c_k e^{i2\pi k t/T}$, $c_k = \frac{a_k - ib_k}{2}$, $c_{-k} = \frac{a_k + ib_k}{2}$; $c_k = \frac{1}{T} \int_{-T/2}^{T/2} g(t) e^{-i2\pi k t/T} dt$; $c_0 = a_0$. ***Pointwise equality stated at continuity points.***

DFT / inverse / FFT

$\omega = e^{i2\pi/n}$. For $j, k = 0, \dots, n - 1$: $\hat{y}_k = \sum_{j=0}^{n-1} y_j e^{-i2\pi j k/n} = \sum_{j=0}^{n-1} y_j \omega^{-jk}$; $y_j = \frac{1}{n} \sum_{k=0}^{n-1} \hat{y}_k e^{i2\pi j k/n}$. Direct $O(n^2)$; FFT $O(n \log_2 n)$. $y_j \in \mathbb{R}$: $\hat{y}_{n-k} = \overline{\hat{y}_k}$, $\hat{y}_0 \in \mathbb{R}$, $\hat{y}_{n/2} \in \mathbb{R}$ (n even).

Fourier coefficients from equally spaced data

$t_j = j\Delta t$, $\Delta t = T/n$, $y_j \approx g(t_j)$, $p = \lfloor n/2 \rfloor$. $A_0 = \frac{\hat{y}_0}{n}$, $A_k = \frac{2\text{Re}(\hat{y}_k)}{n}$, $B_k = -\frac{2\text{Im}(\hat{y}_k)}{n}$. If $n = 2p$ even: $k = 1, \dots, p - 1$ use ordinary form; $A_p = \hat{y}_p/n$, $B_p = 0$. If $n = 2p + 1$ odd: ordinary form through $k = p$. Then $y_j = A_0 + \sum_k [A_k \cos(2\pi k t_j/T) + B_k \sin(2\pi k t_j/T)]$. **De-trending and spectrum** For $Y_i \approx x_1 + x_2 t_i + g(t_i)$, g periodic: least-squares fit $x_1 + x_2 t$ and define $y_i = Y_i - (x_1 + x_2 t_i)$ before DFT. Power spectrum $p_k = \frac{|\hat{y}_k|^2}{n}$ plotted against frequency k/T ; inspect $k = 0, \dots, \lfloor n/2 \rfloor$ for real data. Large p_k identify dominant periodic components; period = T/k (***$k > 0$***).

7. IMPLIED VOLATILITY & NONLINEAR EQUATIONS

Black-Scholes call and implied volatility
 $\tau = T - t > 0$. $c = SN(d_1) - Ke^{-r\tau}N(d_2)$;
 $d_1 = \frac{\ln(S/K) + r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau}$, $d_2 = d_1 - \sigma\sqrt{\tau}$; $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\xi^2/2} d\xi$.
Put-call parity: $c + Ke^{-r\tau} = p + S$.
Require $S > 0$, $K > 0$, $\sigma > 0$, $\tau > 0$; model assumptions in Topic 10.
Implied volatility: $f(\sigma) = c(\sigma) - c_{\text{mkt}} = 0$, $f'(\sigma) = SN'(d_1)\sqrt{\tau} > 0$.
Existence and uniqueness when $\max(S - Ke^{-r\tau}, 0) \leq c_{\text{mkt}} \leq S$.
 $f(0^+) = \max(S - Ke^{-r\tau}, 0) - c_{\text{mkt}}$, $f(\infty) = S - c_{\text{mkt}}$.

Taylor and finite differences
If $f \in C^{n+1}[a, b]$, $x, x+h \in [a, b]$: $f(x+h) = \sum_{k=0}^n \frac{h^k f^{(k)}(x)}{k!} + O(h^{n+1})$.

$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) = \frac{f(x) - f(x-h)}{h} + O(h)$ **Require $f \in C^2$ and shifted point in $[a, b]$.**
 $f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$ **Require $f \in C^3$ and $x \pm h \in [a, b]$.**
 $f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2)$ **Require $f \in C^4$ and $x \pm h \in [a, b]$.**
Machine-rounding balance:
 $O(h) + O(\epsilon h) : h \asymp e^{1/2}$, $O(h^2) + O(\epsilon h) : h \asymp e^{1/3}$, $O(h^2) + O(\epsilon h^2) : h \asymp e^{1/4}$.
 h must be representable so $x \pm h$ differs from x .

Scalar iterations
 $e^{(k)} = x^{(k)} - x^*$, $\beta = \lim_{k \rightarrow \infty} \frac{e^{(k+1)}}{(e^{(k)})^p}$; for $v=1$ require $|\beta| < 1$.

Newton: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$. Quadratic locally; **require $f(x^*)=0$, f and f' continuous near x^* , $f' \neq 0$ there, initial guess sufficiently close**; asymptotic ratio $\rightarrow |f''(x^*)|/(2|f'(x^*)|)$.
Secant: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{\frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}}$; order $(1+\sqrt{5})/2$ locally.
Require two distinct initial iterates, nonzero divided differences, starts sufficiently close to a simple root.

Systems / Gauss-Newton
 $r: \mathbb{R}^n \rightarrow \mathbb{R}^m$; $J_{ij} = \frac{\partial r_i}{\partial x_j}$, $r(x+d) = r(x) + J(x)d + O(\|d\|^2)$. For $m=n$,
Newton-Raphson: $J(x^{(k)})d^{(k)} = -r(x^{(k)})$, $x^{(k+1)} = x^{(k)} + d^{(k)}$; **J must be nonsingular at each solve; local quadratic convergence under smoothness and a sufficiently close start.**
For $m \geq n$ nonlinear least squares $\min \|r\|_2^2$: $J^T J d = -J^T r$.
Local step is uniquely defined when J has full column rank.
 $J_{ij} \approx \frac{r_i(x+he_j) - r_i(x)}{h}$, $J_{ij} \approx \frac{r_i(x+he_j) - r_i(x-he_j)}{2h}$; use $h \approx \epsilon^{1/2}$ forward, $h \approx \epsilon^{1/3}$ central.

8. NUMERICAL INTEGRATION

Probability integrals and quadrature
PDF p : $p \geq 0$, $\int \mathbb{R} p = 1$. $E[f(X)] = \int_{-\infty}^{\infty} f(x)p(x) dx$, $P(x) = \int_{-\infty}^x p(\xi) d\xi$.
If P is differentiable and strictly increasing, $y = P(x)$ gives $E[f(X)] = \int_0^1 f(P^{-1}(y)) dy$.
 $I(f) = \int_a^b f(x) dx$, $Q(f) = \sum w_i f(\xi_i)$, $E = I - Q$. Degree of
precision m : exact for all polynomials degree $\leq m$ and not all of degree $m+1$.
Composite rules
Grid $a = x_0 < \dots < x_N = b$, $h_i = x_i - x_{i-1}$. Trapezoidal:
 $Q_N^{\text{trap}} = \sum \frac{h_i}{2} (f_{i-1} + f_i)$. Uniform $h = (b-a)/N$:
 $h[1/2 f_0 + \sum_{i=1}^{N-1} f_i + 1/2 f_N]$.
Midpoint: $Q_N^{\text{mid}} = \sum_{i=1}^N h_i f(\frac{x_{i-1} + x_i}{2})$.

$|E_N^{\text{trap}}| \leq \frac{(b-a)Mh^2}{12}$, $|E_N^{\text{mid}}| \leq \frac{(b-a)Mh^2}{24}$. **Require $f \in C^2[a, b]$ for $O(h^2)$.** Both have degree 1.
Composite Simpson, **uniform grid and even N** :
 $Q_N^S = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + \dots + 4f_{N-1} + f_N]$. Degree 3; $|E_N^S| \leq \frac{(b-a)\max|f^{(4)}| h^4}{2880}$.
Require $f \in C^4[a, b]$ for $O(h^4)$.
 $|E_{2N}| \leq \frac{|E_N|}{2^2}$, $r \approx \log_2 \frac{|E_N|}{|E_{2N}|}$.

Gauss-Legendre
On $[-1, 1]$, N nodes/weights chosen exact through x^{2N-1} ; degree $2N-1$.
 $N=2$: $\xi = \pm \frac{1}{\sqrt{3}}$, $w=1$; $N=3$: $\xi = 0, \pm \sqrt{\frac{3}{5}}$, $w = \frac{8}{9}, \frac{5}{9}, \frac{5}{9}$.
 $\int_a^b f(x) dx \approx \frac{b-a}{2} \sum_{i=1}^N w_i f(\frac{a+b}{2} + \frac{b-a}{2} \xi_i)$.
Non-smooth and multidimensional
Formal orders are not guaranteed when required derivatives are missing/unbounded. Split at nonsmooth points or change variables to regularise before applying a high-order rule.

Tensor product: $\int_{[0,1]^d} f(x, y) dx dy \approx \sum_{i,j} w_i w_j f(\xi_i, \xi_j)$. A d -fold
($N+1$)-point rule uses $(N+1)^d$ points (**curse of dimensionality**).

9. RANDOM NUMBERS & SIMULATION — DISTRIBUTIONS

True random: physical source. Pseudorandom: deterministic, reproducible, statistically random-like. Quasirandom: deterministic, designed for multidimensional uniformity.
 $X \sim U[a, b]$: $p(x) = \frac{1}{b-a}$, $E[X] = \frac{a+b}{2}$, $\text{Var}(X) = \frac{(b-a)^2}{12}$ (**$a < b$**); if
 $U \sim U[0, 1]$, $Y = a + (b-a)U$.
 $U \sim U[0, 1]$, $Y = F^{-1}(U) \implies F_Y = F$. **Use a suitable generalised inverse if strict invertibility fails.**
 $Y \sim N(\mu, \sigma^2)$: $p(y) = \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi\sigma}}$, $Y = \mu + \sigma Z$ (**$\sigma > 0$**). Independent
sums: $\Sigma EX_i = \Sigma EX_i$; $\text{Var} \Sigma X_i = \Sigma \text{Var} X_i$. CLT: iid sums are approximately normal for large n and finite variance.

9. RANDOM NUMBERS & SIMULATION — MC/QMC

Monte Carlo
 $E[f] = \int_{\mathbb{R}} f(x)p(x) dx$, $x_i \stackrel{\text{iid}}{\sim} p$
 $E_N[f] = \frac{1}{N} \sum_{i=1}^N f(x_i)$
Estimator is unbiased; RMS error is independent of d .
 $e_N[f] = \frac{\sigma[f]}{\sqrt{N}} = O(N^{-1/2})$
 $e_N[f] \leq \text{tol} \implies N \geq \left(\frac{\sigma[f]}{\text{tol}}\right)^2$

Require independent samples and finite $\text{Var}[f]$. Variance reduction
Control variate: known $\mu_g = E[g]$, estimate $E[f-g] + \mu_g$; useful when $\text{Var}(f-g) \ll \text{Var}(f)$.
Antithetic pairs: **N even**; on symmetric $\mathbb{R}^d$ pair x and $-x$; on $[0, 1]^d$ pair x and $1-x$. Odd component integrates exactly.

Importance sampling: $q(x) = g(x)p(x)$, $E[f] \approx \frac{1}{N} \sum_{i=1}^N \frac{f(y_i)}{g(y_i)}$, $y_i \sim q$.
Require $q \geq 0$, $\int q = 1$, and $g(x) \neq 0$ wherever $f(x) \neq 0$; finite variance of f/g under q .

Quasi-Monte Carlo
Deterministic points:
 $Q_N = \frac{1}{N} \sum_{j=1}^N f(x_j)$, $x_j \in [0, 1]^d$
Koksma-Hlawka:
 $|\int_{[0,1]^d} f - Q_N| \leq V(f) D_N^*$
Requires bounded variation $V(f)$. Star discrepancy:

$$D_N^* = \sup_{y \in [0,1]^d} \left| \prod_{i=1}^d y_i - \frac{\#\{x_j \in [0, y]\}}{N} \right|, \quad D_N^* \leq C \frac{(\log N)^d}{N}$$

10. STOCHASTIC INTEGRATION & SDEs — FOUNDATIONS

Random walk / Wiener process
 $t_n = n\Delta t$, $\Delta t = T/N$:
 $\Delta X_n = \mu \Delta t + \sigma(\Delta t)^{\gamma} Z_n$, $E[X_N - X_0] = \mu T$, $\text{Var}(X_N - X_0) = \sigma^2 T(\Delta t)^{2\gamma-1}$;
iid $Z_n \sim N(0, 1)$. Finite nonzero limit requires $\gamma = \frac{1}{2}$.
 $S_{n+1} = S_n \left(1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_n\right)$.
Wiener W: $W(0)=0$; a.s. continuous paths; increments over non-overlapping intervals are independent: for $a < b \leq c < d$, $W(b)-W(a)$ is independent of $W(d)-W(c)$;
 $W(t) - W(s) \sim N(0, t-s)$, $W_{j+1} = W_j + \sqrt{\Delta t} Z_j$.
Itô integral / lemma
 $\int_0^T f(W(t), t) dW(t) = \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} f(W(t_n), t_n) [W(t_{n+1}) - W(t_n)]$ (**left endpoints; adapted integrand**).
 $\sum (\Delta W_n)^2 \rightarrow T$, $\int_0^T W dW = \frac{1}{2} W(T)^2 - \frac{1}{2} T$.
 $dX = f(X, t) dt + g(X, t) dW$, $X(0) = X_0$. For $Y = H(X, t)$:
 $dY = (H_t + \frac{1}{2} g^2 H_{xx}) dt + H_x dX$. **Require H continuously differentiable in t and twice continuously differentiable in x .**

10. STOCHASTIC INTEGRATION & BLACK-SCHOLES

Geometric Brownian motion
 $dS/S = \mu dt + \sigma dW$, **$S_0 > 0$, $\sigma > 0$** . $S(t) = S_0 \exp\left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W(t)\right]$;
 $\ln\left(\frac{S(t)}{S_0}\right) \sim N\left((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t\right)$; $S(t)$ is lognormal and positive.
Black-Scholes PDE and conditions
 $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$, $S > 0$, $0 < t \leq T$. Delta hedge uses $f = V_S$ short shares. **Assumptions: GBM underlying; constant known r, σ ; no dividends, transaction costs, taxes or arbitrage; continuous trading and short selling allowed.**
 $c(S, T) = \max(S - K, 0)$, $c(0, t) = 0$, $c \sim S - Ke^{-r(T-t)}$, $p(S, T) = \max(K - S, 0)$, $p(0, t) = Ke^{-r(T-t)}$, $p \rightarrow 0$.
Put-call parity: $S + p = c + Ke^{-r(T-t)}$ (**European options, same K, T**).

Log-price/time transformation
 $y = \ln(S/K)$, $\tau = T - t$, $V(S, t) = U(y, \tau)$:
 $-U_\tau + (r - \frac{1}{2}\sigma^2)U_y + \frac{1}{2}\sigma^2 U_{yy} = rU$; $y \in \mathbb{R}$, $\tau \in [0, T]$.
 $U(y, 0) = K \max(e^y - 1, 0)$, $U \rightarrow 0$ ($y \rightarrow -\infty$), $U \sim Ke^y - Ke^{-r\tau}$ ($y \rightarrow \infty$);
 $U(y, 0) = K \max(1 - e^y, 0)$, $U \sim Ke^{-r\tau}$ ($y \rightarrow -\infty$), $U \rightarrow 0$ ($y \rightarrow \infty$).

11. FDMs FOR BLACK-SCHOLES

Grid and differences
Truncate $S \in [0, S_{\max}]$, **$S_{\max} > K$ and sufficiently large**.
 $\Delta S = \frac{S_{\max}}{N_S + 1}$, $S_j = j\Delta S$; $\Delta t = \frac{T}{N_t}$, $t_n = n\Delta t$.
 $V_S \approx \frac{V_{j+1} - V_{j-1}}{2\Delta S}$, $V_{SS} \approx \frac{V_{j+1} - 2V_j + V_{j-1}}{(\Delta S)^2}$; each $O((\Delta S)^2)$. Time
one-sided error $O(\Delta t)$. Boundary/terminal values from Topic 10.
Backward Euler - EXPLICIT scheme
 $\rho = \frac{\Delta t}{2(\Delta S)^2}$, $\alpha_j = (\sigma^2 S_j^2 - rS_j\Delta S)\rho$, $\beta_j = 1 - r\Delta t - 2\sigma^2 S_j^2 \rho$, $\gamma_j = (\sigma^2 S_j^2 + rS_j\Delta S)\rho$.
 $V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n$. Accuracy $O(\Delta t + (\Delta S)^2)$ when stable.

Sufficient stability: $\sigma^2 \geq r$, $\frac{\Delta t}{(\Delta S)^2} \leq \min_{1 \leq j \leq N_S} \frac{1}{2\sigma^2 S_j^2 + \sigma^2 S_j^2}$; then
 $\alpha, \beta, \gamma \geq 0$ and $|\alpha| + |\beta| + |\gamma| \leq 1$. Under this mesh relation overall order is $O((\Delta S)^2)$.
Forward Euler - IMPLICIT scheme
 $\alpha_j = \frac{j\Delta t}{2}(r - \sigma^2 j)$, $\beta_j = 1 + r\Delta t + \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2}$, $\gamma_j = -\frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2} - \frac{rS_j \Delta t}{2\Delta S}$. Solve
tridiagonal $\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1}$ with boundary terms
in RHS, backwards $N = N_t - 1, \dots, 0$. Accuracy $O(\Delta t + (\Delta S)^2)$;
unconditionally stable. **$\sigma^2 \geq r$ is sufficient for strict diagonal dominance/nonsingularity, not necessary.**
Factor once if r, σ time-independent; total $O(N_t N_S)$.

Crank-Nicolson - IMPLICIT scheme; Greeks
At $t_{n+1/2}$: $V_j^{n+1/2} = \frac{V_j^n + V_j^{n+1}}{2}$, $V_t \approx \frac{V_j^{n+1} - V_j^n}{\Delta t}$; use midpoint value
in all S terms. Implicit; accuracy $O((\Delta t)^2 + (\Delta S)^2)$.
 $\Delta = V_S$, $\Gamma = V_{SS}$, $\Theta = V_t$, $\rho = V_r$, $\text{Vega} = V_\sigma$.